Problem 04: Crop Types Along the Mighty River Indus

Time limit: 2 seconds

The Indus River is the seventh-longest river in Asia, with a length of approximately 3,180 km (1,980 mi). Along River Indus, which is by far the largest river in Pakistan, the seventh largest in Asia and among the 50 largest rivers by annual flow, the farmlands are arranged in a straight line. Each farm grows exactly one type of crop. The crops are represented by integers, where different integers correspond to different crop types.
The Government of Khyber Pakhtunkhwa wants to estimate the number and type of crops which are planted by their farmers so that they can plan for import/export of food items in the next cycle. They are particularly interested in identifying the dominant crop, and have sought help from you.
You are given the crop type of each farm in order from north to south of the mighty river Indus. A crop type x is said to be dominant in a segment of consecutive farms **[*L*, *R*]** if x appears strictly more than half of the farms in that segment.

Your task is to process several queries. Each query gives a range of farms **[*L*, *R*]**, and you must determine whether there is any crop type that is dominant in that segment. If such a crop exists, output its type. If no crop is dominant, output 0.

Input:

The input consists of:

- A line containing two space-separated integers N and Q $(1 \le N, Q \le 200,000)$, the number of farms and number of queries.
- A line containing N integers $c_1, c_2, \ldots, c_N$ representing the crop type of each farm. Crop types satisfy $|c_i| \le 10^9$.
- Q lines follow. Each line contains two integers L and R $(1 <= L, R <= N)$, describing a query segment.

Output:

For each query **[L, R],** output a single integer:

- the type of the dominant crop in that segment, if it exists; or
- **0** if no crop type is dominant.

Sample Input	Sample Output
7 5 1 2 2 2 3 2 1 1 7 2 6 3 5 1 3 5 7	2 2 2 2 0

Sample Explanation:

- In the segment [1,7], the crop type *2* appears 4 times, which is more than half of 7.
- In the segment [5,7] = [3,2,1], no crop appears more than once; length is 3, so no crop is dominant.